Sahaj Bhadja

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Summary

Forward-deployed software engineer with 2+ years building **healthcare dashboards**, SQL/ETL workflows, APIs, and AI-assisted tools; strong at translating messy operations problems into validated software.

Education

University of Southern California, Los Angeles, CA — **August 2024 – May 2026**
Master of Science in Computer Science — *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India — **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering — *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife — **Los Angeles, CA**
Software Engineer — *September 2025 – Present*

- Built **operations dashboards and platform workflows** for 500+ users, translating stakeholder pain points into APIs, SQL models, validation logic, monitored jobs, release notes, support runbooks, and reviewable handoffs.
- Owned **data quality and performance checks** across indexed queries, background jobs, and CI/CD releases, reducing dashboard latency from **1.2s to 350ms** with traceable production handoffs and QA notes.

Grad8 — **Los Angeles, CA**
Software Engineer Intern — *June 2025 – August 2025*

- Developed **React interfaces and Flask APIs** handling 10,000+ daily requests, implementing file ingestion, form workflows, automated checks, documented fixes, sprint-ready production releases, business-facing dashboards, and reports.
- Verified **AI-generated grading outputs** by reproducing OCR, persistence, and execution defects, documenting regression cases, reviewing fixes, and cutting instructor feedback turnaround by **60%** across validation cycles and QA notes.

USC AD Labs — **Los Angeles, CA**
ML Research Assistant — *January 2025 – May 2025*

- Created **Python evaluation pipelines** across 100,000+ records, producing reproducible metrics, logs, reports, and review artifacts for validating model behavior and data-driven decisions under experiment constraints and audits.
- Documented **edge cases and system behavior** for researchers, turning ad hoc experiments into repeatable QA scenarios with traceable outputs, clearer handoffs, and reviewable conclusions for implementation planning.

Aasma Technology Solutions — **India**
Software Developer Engineer - (SecureSync) — *August 2023 – July 2024*

- Engineered **SQL and ETL services** using Python, AWS Glue, and S3, improving response time by **30%** across integrations, releases, reviews, business handoffs, analytics workflows, reporting, and summaries.
- Maintained **20+ automated test suites** using Jenkins, PyTest, and Mockito across validation flows, increasing CI reliability by **68%** and reducing release regressions during coordinated deployments and reviews.

Aasma Technology Solutions — **India**
Full Stack Developer Intern - (VibrantChem) — *May 2023 – August 2023*

- Built **CRM-style application modules** with REST APIs, WebSockets, and SQL/NoSQL persistence, supporting 2M+ synchronized records across dashboards, workflows, integrations, code reviews, releases, support needs, and practice logs.
- Automated **Docker-based build and test flows** with Jenkins, cutting release time by **50%** while improving deployment consistency, troubleshooting documentation, and shared-environment handoffs across staging workflows and release checks.

Skills

Languages:	TypeScript, JavaScript, Python, SQL, Java, C++, Bash
Frontend:	React, Vite familiarity, Tailwind CSS, dashboards, forms, reports, business-facing workflows
Backend:	Express familiarity, Flask, FastAPI, REST APIs, service design, validation, code review, Git
Data/ETL:	PostgreSQL, MySQL, SQLite familiarity, DuckDB familiarity, SQL modeling, ETL, analytics pipelines
AI/Delivery:	AI-assisted coding, output verification, QA gates, Docker, GitHub Actions, CI/CD, Agile, Lean interest

Projects

HMIS Live | *MERN, TypeScript, ABDM APIs, Sentry*

- Built a **healthcare operations dashboard** with patient workflows, consent-based history access, appointment scheduling, Twilio alerts, Sentry monitoring, and secure TypeScript APIs, reducing administrative workload by **30%**.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Built an **AI-powered SQL workspace** supporting 10+ databases, accelerating query authoring, schema exploration, and data validation through Java services and TypeScript interfaces with Docker workflows and reviewable outputs.

FinNexus Live | *Next.js, TypeScript, MongoDB, WebSockets*

- Built a **secure analytics dashboard** supporting 50K+ users and 1M+ transactions with token-based access, real-time updates, CI/CD-backed deployments, monitoring, audit-friendly workflows, operational reporting, review logs, and summaries.